

Week-1:

Program:

```
CREATE DATABASE IF NOT EXISTS bookflow_db;
```

Output:

#	Time	Action	Message	Duration / Fetch
✓ 1	12:00:54	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected	0.141 sec

Program:

```
USE bookflow_db;
```

Output:

✓ 2	12:01:28	USE bookflow_db	0 row(s) affected	0.015 sec
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Program:

```
CREATE TABLE books(  
  book_id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255) NOT NULL,  
  isbn VARCHAR(13) NOT NULL UNIQUE,  
  published_year INT,  
  CONSTRAINT chk_published_year CHECK(published_year < 2027)  
);
```

Output:

✓ 3	12:03:44	CREATE TABLE books(book_id INT AUTO_INCREMENT PRIMAR...	0 row(s) affected	0.078 sec
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Program:

```
CREATE TABLE members(  
  member_id INT AUTO_INCREMENT PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(160) NOT NULL UNIQUE  
);
```

Output:

✓ 4	12:04:27	CREATE TABLE members(member_id INT AUTO_INCREMENT PRI...	0 row(s) affected	0.093 sec
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Program:

```
DESCRIBE books;
```

Output:

✓ 5	12:05:07	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
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Result Grid						
Filter Rows:		Export:  Wrap Cell Content: 				
	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	

Program:

```
DESCRIBE members;
```

Output:

✓	6	12:05:59	DESCRIBE members	3 row(s) returned	0.016 sec / 0.000 sec
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Result Grid						
Filter Rows:		Export:  Wrap Cell Content: 				
	Field	Type	Null	Key	Default	Extra
▶	member_id	int	NO	PRI	NULL	auto_increment
	full_name	varchar(100)	NO		NULL	
	email	varchar(160)	NO	UNI	NULL	

Program:

```
INSERT INTO books(title,isbn,published_year) VALUES
('The Alchemist','9780061122415','1988'),
('clean code','9780132350884','2008'),
('Atomic Habits','9780735211292','2018');
```

Output:

✓	7	12:06:50	INSERT INTO books(title,isbn,published_year) VALUES ('The Alch...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
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Program:

```
INSERT INTO members(full_name,email) VALUES
('SAHASRA ','sahasra@gmail.com'),
('TEJASWINI','teju@gmail.com'),
('PAVANI ','pavs@gmail.com');
```

Output:

✓	8	12:08:00	INSERT INTO members(full_name,email) VALUES ('SAHASRA ','s...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
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Program:

```
SELECT * FROM books;
```

Output:

9 12:08:49 SELECT * FROM books LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	clean code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM members;
```

Output:

10 12:10:09 SELECT * FROM members LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

	member_id	full_name	email
▶	1	SAHASRA	sahasra@gmail.com
	2	TEJASWINI	teju@gmail.com
	3	PAVANI	pavs@gmail.com
*	NULL	NULL	NULL

Program:

```
> CREATE TABLE student(  
    student_id VARCHAR(100) NOT NULL UNIQUE,  
    student_name VARCHAR(50) NOT NULL,  
    branch VARCHAR(20) NOT NULL  
);
```

Output:

11 12:10:48 CREATE TABLE student(student_id VARCHAR(100) NOT NUL... 0 row(s) affected 0.046 sec

Program:

```
INSERT INTO student(student_id, student_name, branch)
VALUES
('2520030008', 'SAHASRA', 'CSE'),
('2520030011', 'TEJASWINI', 'CSE');
```

Output:

✓ 13 12:12:19 INSERT INTO student(student_id, student_name, branch) VALUE... 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 0.032 sec

Program:

```
SELECT * FROM student;
```

Output:

✓ 14 12:13:00 SELECT * FROM student LIMIT 0, 1000 2 row(s) returned 0.000 sec / 0.000 sec

Result Grid			
Filter Rows:			
	student_id	student_name	branch
▶	2520030008	SAHASRA	CSE
	2520030011	TEJASWINI	CSE
✱	NULL	NULL	NULL

Program:

```
CREATE DATABASE IF NOT EXISTS std_marks;
```

Output:

✓ 1 12:17:36 CREATE DATABASE IF NOT EXISTS std_marks 1 row(s) affected 0.000 sec

Program:

```
USE std_marks;
```

```
CREATE TABLE students_marks1(  
roll_no INT PRIMARY KEY,  
student_name VARCHAR(50),  
std_subject VARCHAR(50),  
marks DECIMAL(5,2));
```

Output:

✓ 2 12:18:28 USE std_marks 0 row(s) affected 0.016 sec
✓ 3 12:18:29 CREATE TABLE students_marks1(roll_no INT PRIMARY KEY, stude... 0 row(s) affected 0.062 sec

Program:

```
DESCRIBE students_marks1;
```

Output:

✓ 4 12:19:06 DESCRIBE students_marks1 4 row(s) returned 0.000 sec / 0.000 sec

Result Grid						
		Filter Rows:			Export:	Wrap Cell
	Field	Type	Null	Key	Default	Extra
▶	roll_no	int	NO	PRI	NULL	
	student_name	varchar(50)	YES		NULL	
	std_subject	varchar(50)	YES		NULL	
	marks	decimal(5,2)	YES		NULL	

Program:

```
INSERT INTO students_marks1(roll_no,student_name,std_subject,marks) VALUES
(1, 'Ravi', 'Math', 85.50),
(2, 'Sita', 'Math', 92.75),
(3, 'Anil', 'Math', 78.40),
(4, 'Priya', 'Math', 88.90),
(5, 'Vijay', 'Math', 80.25),
(6, 'subbusir', 'aws', 98.50),
(7, 'dwaraka', 'dbms', 95.75),
(8, 'ranjani', 'english', 97.40),
(9, 'kavithamam', 'aws1', 99.90),
(10, 'seetha', 'azure', 82.25);
```

Output:

5 12:19:54 INSERT INTO students_marks1(roll_no,student_name,std_subject,m... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 0.031 sec

Program:

```
SELECT *FROM students_marks1;
```

Output:

6 12:20:44 SELECT *FROM students_marks1 LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec

Result Grid				
Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	3	Anil	Math	78.40
	4	Priya	Math	88.90
	5	Vijay	Math	80.25
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
	10	seetha	azure	82.25
•	NULL	NULL	NULL	NULL

Program:

```
SELECT COUNT(*) AS total_students FROM students_marks1;
```

Output:

7 12:21:35 SELECT COUNT(*) AS total_students FROM students_marks1 LIM... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	
	total_students
▶	10

Program:

```
SELECT SUM(marks) FROM students_marks1;
```

Output:

✓	8	12:23:50	SELECT SUM(marks) FROM students_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
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Result Grid	
	SUM(marks)
▶	899.60

Program:

```
SELECT AVG(marks) FROM students_marks1;
```

Output:

✓	9	12:24:36	SELECT AVG(marks) FROM students_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
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Result Grid	
	AVG(marks)
▶	89.960000

Program:

```
SELECT MAX(marks) FROM students_marks1;
```

Output:

✓	10	12:26:25	SELECT MAX(marks) FROM students_marks1 LIMIT 0, 1000	1 row(s) returned	0.031 sec / 0.000 sec
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Result Grid	
	MAX(marks)
▶	99.90

Program:

```
SELECT MIN(marks) FROM students_marks1;
```

Output:

✓	11	12:30:25	SELECT MIN(marks) FROM students_marks1 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
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Result Grid	
	MIN(marks)
▶	78.40

Program:

```
SELECT * FROM students_marks1 WHERE marks > 85;
```

Output:

✓	12	12:31:22	SELECT * FROM students_marks1 WHERE marks > 85 LIMIT 0, 1... 7 row(s) returned	0.016 sec / 0.000 sec
Result Grid		Filter Rows:		
	roll_no	student_name	std_subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	4	Priya	Math	88.90
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 WHERE marks >= 90;
```

Output:

✓	13	12:32:15	SELECT * FROM students_marks1 WHERE marks >= 90 LIMIT 0, ... 5 row(s) returned	0.000 sec / 0.000 sec
Result Grid		Filter Rows:		
	roll_no	student_name	std_subject	marks
▶	2	Sita	Math	92.75
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 WHERE marks < 80;
```

Output:

✓	14	12:32:59	SELECT * FROM students_marks1 WHERE marks < 80 LIMIT 0, 1... 1 row(s) returned	0.000 sec / 0.000 sec
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Result Grid   Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	3	Anil	Math	78.40
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 WHERE marks BETWEEN 80 AND 90;
```

Output:

✓ 15 12:33:54 SELECT * FROM students_marks1 WHERE marks BETWEEN 80 ... 4 row(s) returned 0.000 sec / 0.000 sec

Result Grid   Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	5	Vijay	Math	80.25
	10	seetha	azure	82.25
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 WHERE student_name LIKE 'P%';
```

Output:

✓ 16 12:34:38 SELECT * FROM students_marks1 WHERE student_name LIKE '...' 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid   Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	4	Priya	Math	88.90
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 WHERE student_name IN ('Ravi', 'Sita', 'Vijay');
```

Output:

✓ 17 12:35:22 SELECT * FROM students_marks1 WHERE student_name IN (Ra... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid				
Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	5	Vijay	Math	80.25
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 WHERE marks > 85 AND (std_subject = 'Math' OR student_name LIKE 'P%');
```

Output:

18 12:37:12 SELECT * FROM students_marks1 WHERE marks > 85 AND (std_subject = 'Math' OR student_name LIKE 'P...' 3 row(s) returned 0.015 sec / 0.000 sec				
	roll_no	student_name	std_subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	4	Priya	Math	88.90
✱	NULL	NULL	NULL	NULL

Program:

```
UPDATE students_marks1 SET marks = 90.00 WHERE roll_no = 3;
```

Output:

19 12:37:45 UPDATE students_marks1 SET marks = 90.00 WHERE roll_no = 3 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.015 sec				
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Program:

```
UPDATE students_marks1 SET std_subject = 'Remedial Math' WHERE marks < 80;
```

Output:

20 12:38:51 UPDATE students_marks1 SET std_subject = 'Remedial Math' W... 0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0 0.000 sec				
--	--	--	--	--

Program:

```
DELETE FROM students_marks1 WHERE roll_no = 5;
```

Output:

21 12:39:27 DELETE FROM students_marks1 WHERE roll_no = 5 1 row(s) affected 0.047 sec				
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Program:

```
DELETE FROM students_marks1 WHERE marks < 75;
```

Output:

22 12:40:02 DELETE FROM students_marks1 WHERE marks < 75 0 row(s) affected 0.000 sec				
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Program:

```
SELECT * FROM students_marks1 ORDER BY marks ASC;
```

Output:

23 12:41:04 SELECT * FROM students_marks1 ORDER BY marks ASC LIMIT ... 9 row(s) returned

0.000 sec / 0

	roll_no	student_name	std_subject	marks
▶	10	seetha	azure	82.25
	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	3	Anil	Math	90.00
	2	Sita	Math	92.75
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	6	subbusir	aws	98.50
	9	kavithamam	aws1	99.90
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 ORDER BY marks DESC;
```

Output:

24 12:42:07 SELECT * FROM students_marks1 ORDER BY marks DESC LIMIT ... 9 row(s) returned

0.000 sec / 0

	roll_no	student_name	std_subject	marks
▶	9	kavithamam	aws1	99.90
	6	subbusir	aws	98.50
	8	ranjani	english	97.40
	7	dwaraka	dbms	95.75
	2	Sita	Math	92.75
	3	Anil	Math	90.00
	4	Priya	Math	88.90
	1	Ravi	Math	85.50
	10	seetha	azure	82.25
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 ORDER BY student_name ASC;
```

Output:

25 12:43:15 SELECT * FROM students_marks1 ORDER BY student_name AS... 9 row(s) returned

0.000 sec / 0

Result Grid   Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	3	Anil	Math	90.00
	7	dwaraka	dbms	95.75
	9	kavithamam	aws1	99.90
	4	Priya	Math	88.90
	8	ranjani	english	97.40
	1	Ravi	Math	85.50
	10	seetha	azure	82.25
	2	Sita	Math	92.75
	6	subbusir	aws	98.50
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM students_marks1 ORDER BY student_name DESC;
```

Output:

✓ 26 12:44:15 SELECT * FROM students_marks1 ORDER BY student_name DE... 9 row(s) returned 0.015 sec / 0

Result Grid   Filter Rows:				
	roll_no	student_name	std_subject	marks
▶	6	subbusir	aws	98.50
	2	Sita	Math	92.75
	10	seetha	azure	82.25
	1	Ravi	Math	85.50
	8	ranjani	english	97.40
	4	Priya	Math	88.90
	9	kavithamam	aws1	99.90
	7	dwaraka	dbms	95.75
	3	Anil	Math	90.00
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT std subject, SUM(marks) AS total marks FROM students marks1 GROUP BY std subject;
```

Output:

✓ 27 12:45:13 SELECT std_subject, SUM(marks) AS total_marks FROM students_marks1 GROUP BY std_subject LIMIT 0, 1... 6 row(s) returned 0.000 sec / 0.000 sec

Result Grid			Filter Rows:
	std_subject	total_marks	
▶	Math	357.15	
	aws	98.50	
	dbms	95.75	
	english	97.40	
	aws1	99.90	
	azure	82.25	

Program:

```
SELECT std_subject, AVG(marks) AS avg_marks FROM students_marks1 GROUP BY std_subject;
```

Output:

28 12:46:23 SELECT std_subject, AVG(marks) AS avg_marks FROM students_marks1 GROUP BY std_subject LIMIT 0, 1... 6 row(s) returned 0.000 sec / 0.000 sec

Result Grid			Filter Rows:
	std_subject	avg_marks	
▶	Math	89.287500	
	aws	98.500000	
	dbms	95.750000	
	english	97.400000	
	aws1	99.900000	
	azure	82.250000	

Program:

```
SELECT std_subject, COUNT(*) AS num_students FROM students_marks1 GROUP BY std_subject;
```

Output:

29 12:47:11 SELECT std_subject, COUNT(*) AS num_students FROM students_marks1 GROUP BY std_subject LIMIT 0, ... 6 row(s) returned 0.016 sec / 0.000 sec

Result Grid			Filter Rows:
	std_subject	num_students	
▶	Math	4	
	aws	1	
	dbms	1	
	english	1	
	aws1	1	
	azure	1	

Program:

```
SELECT std_subject, AVG(marks) AS avg_marks
```

Output:

30	12:47:57	SELECT std_subject, AVG(marks) AS avg_marks FROM students_marks1 GROUP BY std_subject HAVING A...	4 row(s) returned	0.016 sec / 0.000 sec
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Result Grid			Filter Rows:
	std_subject	avg_marks	
▶	aws	98.500000	
	dbms	95.750000	
	english	97.400000	
	aws1	99.900000	

Program:

```
FROM students_marks1
GROUP BY std_subject
HAVING AVG(marks) > 90;
```

Output:

33	12:49:19	SELECT std_subject, AVG(marks) AS avg_marks FROM students_marks1 GROUP BY std_subject HAVING A...	4 row(s) returned	0.000 sec / 0.000 sec
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Result Grid			Filter Rows:
	std_subject	avg_marks	
▶	aws	98.500000	
	dbms	95.750000	
	english	97.400000	
	aws1	99.900000	

Program:

```
SELECT std_subject, COUNT(*) AS num_students  
FROM students_marks1  
GROUP BY std_subject  
HAVING COUNT(*) > 1;
```

Output:

34 12:50:08 SELECT std_subject, COUNT(*) AS num_students FROM students_marks1 GROUP BY std_subject HAVING ... 1 row(s) returned 0.000 sec / 0.000 sec

	std_subject	num_students
▶	Math	4

